

School of Science and Engineering

Practice Examination — Statistics and Probability

Course Code: STA 2101

Course Title: Statistics and Probability

Section: 4

Instructor: Atanu Shuvam Roy

Time: 1 Hour

Full Marks: 25

Question 1

[6 Marks]

CO1

- (a) Define Statistics and differentiate between Descriptive and Inferential Statistics. [2]
- (b) Mention three ways statistics can help in analyzing social media usage patterns. [3]
- (c) Explain the difference between a population and a sample using an example related to university students. [1]

Question 2

[9 Marks]

CO2

- (a) A mobile app company has 5,000 users. To estimate the average daily time spent on the app, describe how a **Simple Random Sampling** approach could be applied to select 100 users. [2]
- (b) A tech firm has 400 employees in three departments: Software (150), HR (100), Marketing (150). They want to estimate the average monthly working hours using **Stratified Sampling**. Explain how to select samples proportionally. [2]
- (c) Consider this dataset of 28 students representing weekly hours spent on online courses:

Hours = [2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]

Categorize students by learning intensity:

- Low: ≤ 6 hours
- Medium: 7–12 hours
- High: > 12 hours

Perform **Stratified Sampling** by selecting 2 students from each category. [3]

- (d) The students are also divided into four study groups (A, B, C, D). If we randomly select two study groups and include all students in those groups for analysis, what type of sampling is this? Describe how it works. [2]

- (e) Perform **Cluster Sampling** on the above dataset by treating every 7 consecutive students as a cluster. Randomly select 2 clusters and list the students included. [2]

Question 3
CO3

[10 Marks]

- (a) A local bookstore recorded daily sales (in dozens of books) over a month:

Number of Books Sold	Frequency
10–14	3
15–19	5
20–24	12
25–29	9
30–34	6
35–39	3
40–44	2

- Draw a **Histogram** and a **Frequency Polygon** for this data. [3]
- (b) Identify the shape of the distribution and justify your answer. [2]
- (c) Construct an **Ogive (Cumulative Frequency Curve)** and estimate the median number of books sold. [2]
- (d) Another branch of the bookstore has an average of 25 books sold per day with a standard deviation of 4. Compare the variability between both branches. Which is more consistent? [2]
- (e) Five students scored 60, 65, 70, 75, and x on a mini-project. If the mean score is 68, determine the value of x . [1]

— Good Luck —